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Paleomagnetic data from Triassic strata, Zuni uplift, New Mexico: Further evidence of large-magnitude Triassic apparent polar wander of North America

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Abstract. Upper Carnian (~225 Ma) strata of the Bluewater Creek Formation of the Chinle Group of the Colorado Plateau, western New Mexico, contain predominantly reverse polarity characteristic magnetizations of high coercivity and distributed unblocking temperature up to 695°C. Data from a syndepositional structure suggest acquisition of a stable remanence during or soon after deposition. The mean paleomagnetic pole for 13 virtual geomagnetic poles (VGP) accepted (of 17 sites collected) is located at 55.2°N, 87.5°E (A_{95}=6.7°, K=39.7). The angular distance between this pole and the mean of recently published middle and late Norian (~210 Ma) poles from overlying strata is 12.5°, supporting the hypothesis that significant apparent polar wander occurred during Late Triassic time. Furthermore, the Bluewater Creek pole lies to the west (6.1°, angular distance) of the cratonic reference pole for Carnian time supporting the hypothesis of a small Colorado Plateau rotation since the early Mesozoic. Anisian strata of the Moenkopi Formation contain dual polarity magnetizations of high coercivity and distributed unblocking temperatures up to 690°C. The mean VGP of six sites yields is located at 53.1°N, 96.3°E (A_{95}=6.4°, K=109.2), which is similar to previously determined Middle Triassic poles. Triassic paleomagnetic poles from southwest North America from strata in continuous stratigraphic succession, both on and off the Colorado Plateau, are not consistent with the hypothesis of a Late Triassic stand still of the geomagnetic pole.

1. Introduction

Recent estimates of Colorado Plateau (CP) rotation based on direct paleomagnetic pole comparisons have focused on a relatively large data set from Upper Triassic strata, a time interval during which apparent polar wander (APW) of North America has been interpreted to be slow to negligible [*Kent and Witte*, 1993; *Kent and Olsen*, 1997]. Paleomagnetic data for the Newark Supergroup, on which the hypothesis of small to negligible APW is based, include the mid to upper Carnian Stockton and Lockatong formations, the lower to upper Norian Passaic Formation, and overlying Hettangian basalts and sedimentary rocks (an interval from ~228 to 205 Ma). Other paleomagnetic data, however, have been interpreted to indicate that Late Triassic APW of North America is significant. *Kodama et al.* [1994] showed evidence of Late Triassic APW from Mesozoic rocks of northeast North America. Also, a long succession of Triassic strata in the Sangre de Cristo Mountains of east-central New Mexico, that includes Anisian to latest Norian strata of the Moenkopi Formation and the Chinle Group, records ~25° (angular distance) of APW between ~240 and ~208 Ma [*Molina Garza et al.*, 1996].

Although deposition of the Newark Supergroup and Chinle Group was almost entirely contemporaneous, recent estimates of large CP rotation based on Triassic data may be affected by problems in stratigraphic correlation and incomplete sampling of the Triassic segment of the North American APW path. Available poles for the Colorado Plateau, and thus included in the analysis of Triassic APW and CP rotation performed by *Kent and Witte* [1993], are entirely middle to latest Norian (~215-205 Ma). These poles have been obtained from the Owl Rock and the Church Rock formations of the Chinle Group. The Owl Rock overlies lower Norian strata. Because of its stratigraphic position, it is assumed to be of mid-Norian age [*Lucas*, 1993]. The overlying Church Rock is uppermost Norian and possibly Rhaetian. *Kent and Witte* [1993] compared these data with the mean of three cratonic poles assumed to be of the same age as the CP poles: the upper shale pole of *Bazard and Butler* [1991] (lower Norian), the ~214 Ma (mid-Norian) Manicouagan Impact pole, and the pole for the upper Passaic Formation (lower-upper Norian) of *Witte et al.* [1991]. The age assigned to the Passaic pole is

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based on the observation that the Raritan and Jacksonwald localities of *Witte et al.* [1991] include lower Norian strata. Although the age differences between cratonic and CP poles seem minor, only one of the cratonic poles included in this analysis sampled the latest Triassic segment of the North American APW path. Estimates of the duration of the Norian stage vary. *Harland et al.* [1990] estimated a duration of about 14 million years. This and the more recent geologic timescale of *Gradstein et al.* [1994] suggest that the base of the early Norian is about 10 m.y. older than the base of the late Norian. Therefore the comparison of *Kent and Witte* [1993] is based on results for strata of unquestionably different age. A few million years would not affect previous estimates of large CP rotation if Late Triassic APW was indeed negligible, but the age difference between cratonic and CP poles is sufficient to bias a rotation estimate if APW was rapid.

The new pole for the upper Carnian Bluewater Creek Formation reported here provides a datum of crucial importance to (1) test the hypothesis of a Late Triassic stand still in the North American APW path and (2) improve estimates of CP rotation based on Triassic results. This manuscript focuses on the first issue; regardless of the magnitude of CP rotation, paleopoles from rocks on and off the plateau should record similar amounts of APW. From the determination of two new pole positions and reevaluation of previously published data, we suggest that APW was neither small nor slow during the Late Triassic. This is important not only for understanding the kinematics of deformation along the western margin of North America but also in determining the motion of the North American craton during the Mesozoic. In a companion paper [*Molina Garza et al.*, this issue] we argue that the assumption of a large magnitude (≥10°) rotation of the Colorado Plateau is not supported by Late Carboniferous through Late Jurassic paleomagnetic data.

2. Geology and Sampling

The north margin of the Zuni uplift in the eastern CP (Figure 1) is characterized by continuous exposure of Permian and Triassic strata assigned to the San Andres and Moenkopi Formations and to the Chinle Group. The Upper Permian San Andres Formation consists of massive, tan colored, allochem-rich limestone. It is disconformably overlain by purple-gray to red sandstones, siltstones, and mudstones of the Middle Triassic Holbrook Member of the Moenkopi Formation, which in turn is disconformably overlain by the Upper Triassic Chinle Group [*Lucas and Heckert*, 1994]. The Chinle-Moenkopi contact is marked by quartz-pebble conglomerate and mottled, pedogenically altered, strata of the Shinarump Formation.

The Chinle Group in western New Mexico includes, in ascending order, the Shinarump, Bluewater Creek, Petrified Forest, Owl Rock, and Rock Point (Church Rock) formations (Figure 1). The Bluewater Creek Formation (formerly the Monitor Butte member of the Chinle Formation) consists predominantly of quartzo-feldspathic mudstones ranging in color from gray to reddish brown and is less hematitic than strata of the younger formations of the Chinle Group. The Bluewater Creek Formation is less than 100 m thick. It was deposited in lacustrine environments under varying oxidation conditions. Interfingered lenses of micaceous sandstone are common. A prominent, ledge-forming sandstone sequence is locally named the McGaffey member. The Bluewater Creek Formation contains vertebrate fossils of the Adamanian faunachron [*Lucas*, 1993] and fossil plants of the Dinophyton floral zone [*Ash*, 1980], which establish its age as late Carnian.

We collected 27 sites from San Andres (4 sites), Moenkopi (6 sites) and Bluewater Creek (17 sites) strata from localities near Bluewater Lake State Park and near Fort Wingate (Figure 1). At most sites, strata dip gently (2° to 12°) to the north. Sites of Bluewater Creek strata at the Fort Wingate locality were collected from beds informally referred to as Lake Ciniza [*Ash*, 1978]. Six sites have near vertical dips associated with a syndepositional slump. The slump consists of about 10 m of siltstone and fine sandstone layers overlain by horizontal lying mudstones [*Ash*, 1978; *Lucas and Hunt*, 1993]. At each site, 4 to 12 samples were cored using a portable gas-powered drill. Orientation was recorded using a clinometer, magnetic compass, and Sun compass.

Samples were subjected to alternating field (AF) demagnetization up to 110 mT, and progressive thermal demagnetization (10-20 steps) up to 690°C. Principal component analysis was used in the interpretation of demagnetization results. Site mean directions and mean virtual geomagnetic poles (VGPs) were determined assuming both Fisher and Bingham statistical distributions.

3. Paleomagnetic results

3.1. Upper Carnian Bluewater Creek Formation

Our highest quality pole is based on data from the Bluewater Creek Formation. Demagnetization results for these rocks are of varied quality but consistently yield nearly uniform reverse polarity characteristic magnetizations (ChRMs) of high coercivity and distributed unblocking temperatures (Figure 2). About 25% of the samples used in the final calculations, mostly from the Bluewater Creek locality (bw sites), have simple nearly univectorial remanences directed to the origin with maximum unblocking temperatures in excess of 650°C and line fits that yield low (less than 5°) angular dispersions (Figure 2a). These rocks contain nonuniform distributions of hematite pigment as color-banded crossbeds and laminations.

Most samples from the Fort Wingate locality (fw sites) and some from Bluewater Creek lack hematite pigment and have more complex, multivectorial, magnetizations best defined by combining AF and thermal demagnetization (Figures 2b-2c). Demagnetization first removes a low unblocking temperature (< 200°C) and low coercivity (<40 mT) viscous magnetization with steep positive inclination. In some samples, a high-coercivity (>100 mT) but low unblocking temperature component is subsequently removed between about 100° and 300°C (Figures 2b-2d). This component has steep negative inclination and south-southeast directed declinations (Figure 3d).

The ChRM is isolated at temperatures of above 400°C, but above 550°C demagnetization behavior is sometimes erratic (Figure 2d). The linear trajectories defined between the 400° and 550°C steps of the demagnetization sequences may (Figures 2b and 2c) or may not (Figures 2e and 2f) be directed to the origin. Demagnetization results suggest that the signal remaining at temperatures above ~600°C in samples that lack hematite pigment is randomly scattered. Trends of

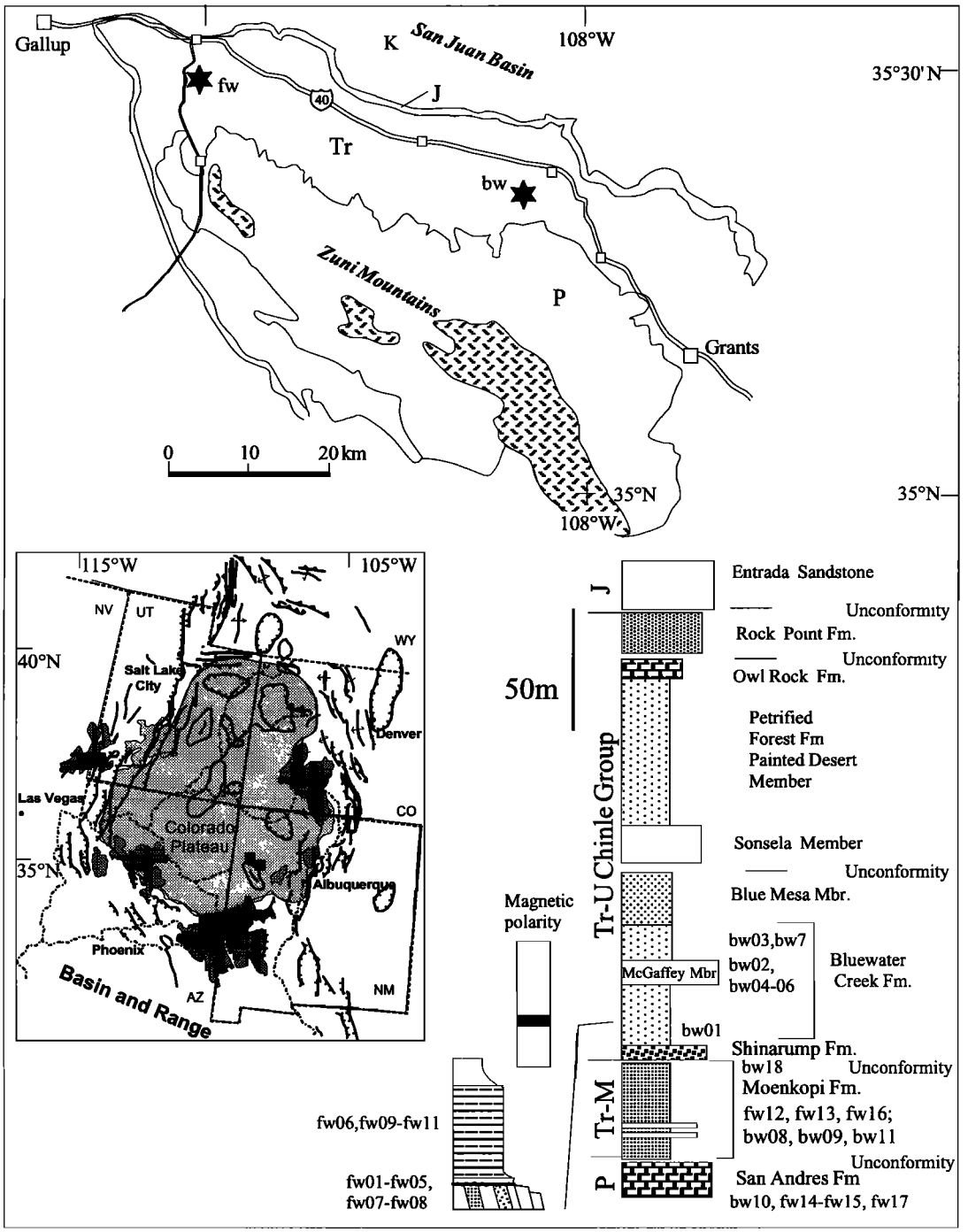


Figure 1. Schematic geologic map of the Zuni Mountains indicating sampling localities (stars). The inset shows a simplified tectonic map of the Colorado plateau (light gray pattern) and surrounding region, showing main structural trends and areas of Tertiary volcanism (dark gray pattern). P, Permian; Tr, Triassic; J, Jurassic; K, Cretaceous; and the stippled pattern is Precambrian basement. A schematic stratigraphic section of the Chinle Group in western New Mexico [after *Lucas and Heckert*, 1994] indicates sampling sites and the interpreted magnetostratigraphy of the Bluewater Creek Formation.

demagnetization trajectories are not consistent, with some endpoints directed to steeper and more south-southwest directions (Figure 2e) and others becoming more shallow and more south-southeast directed (Figure 2f). This suggests that a component of higher unblocking temperature cannot be resolved in these rocks, and measurements above about 600°C probably lack any significance.

For samples with erratic behavior at high temperatures the direction of the ChRM was calculated in two ways. First, line fits were anchored to the origin. In a second calculation, free segments were computed using all points between the 400° and 550°C steps. Results yield indistinguishable site means, typically within a few degrees of one another (Figure 3). If the directions were systematically biased by an unresolved high-

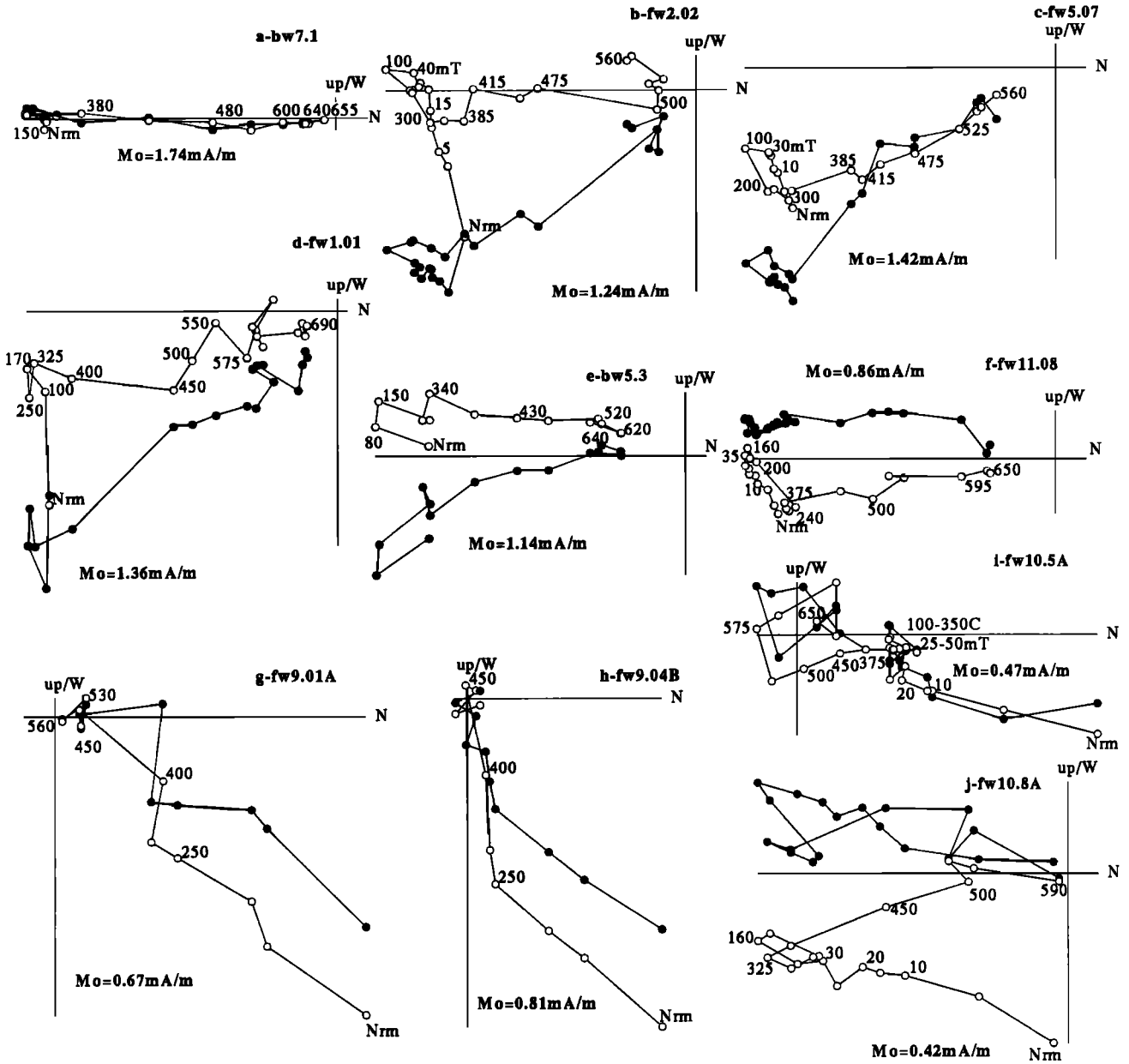


Figure 2. Orthogonal demagnetization diagrams [*Zijderveld*, 1967] of selected specimens of the (a-f) Bluewater Creek Formation, (g-i) Moenkopi Formation, and (j, k) San Andres Formation. All diagrams in in situ coordinates. Open (solid) symbols are projections on the vertical (horizontal) plane. In Figures 2b and 2c, three components of magnetizations are easily identified. Examples in Figures 2g and 2i are interpreted as normal polarity magnetizations.

temperature (>600°) magnetization, we would expect anchored line fits to be systematically displaced toward that direction, but they are not. Results using the second method were used in the final calculations (Table 1). Overall, the average maximum angular deviation (MAD) value for free line segments for samples of the Bluewater Creek Formation is 7.9°, attesting to the quality of the data. Only 18% of the included sample directions have MAD values between 10° and 15°. Samples with MAD values greater than 15° were excluded from all calculations.

In field coordinates, the five sites collected from beds forming part of a syndepositional slump have a clustered group of southeast-directed mean directions that are statistically distinct and about 35° (angular distance) from the rest of the Bluewater Creek sites (Figure 3a and Table 1). Tilt correction restores these directions to within 20° of the mean for shallowly dipping strata (Figure 3b). The tilt correction does not produce a statistically significant fold test [*McFadden and Jones*, 1981]. However, the k parameter increases from an in situ value of 16.6 to a tilt corrected value of 27.3, with a maximum of 29.3 at 59% unfolding. In both in situ and tilt corrected coordinates the means are statistically distinct. We note, however, that in situ site means for sites on the slump have positive inclinations and are far removed from expected Late Triassic or younger directions. Tilt-corrected site means, on the other hand, are consistent with

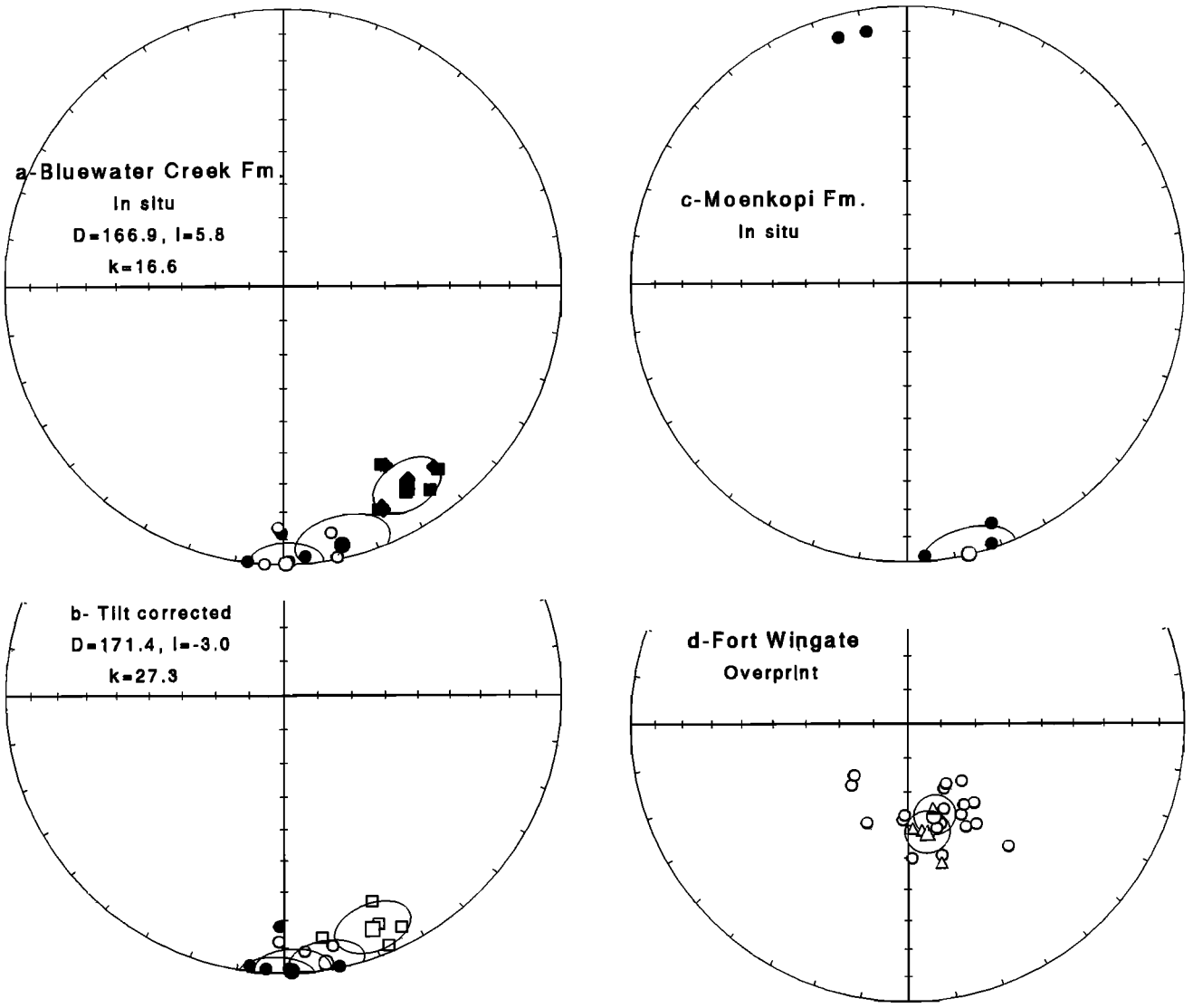


Figure 3. Equal-area projections of site mean directions in the (a-b) Bluewater Creek and (c) Moenkopi formations. Sites collected from a syndepositional slump in the Bluewater Creek section are plotted as squares and diamonds; diamonds are calculations anchoring line fits to the origin which results in nearly identical site means, (see text for explanations). In Figures 3a and 3b, we also show the mean direction of sites from the syndepositional slump and the mean from gently north dipping sites. After tilt correction the angular distance between directions found in slumped and flat lying strata is less than 20°. (d) Directions of magnetizations interpreted as late Mesozoic overprints in Triassic (circles) and Permian (triangles) strata. Open (solid) symbols are lower (upper) hemisphere projections.

paleolatitudes expected for the sampling locality. The fact that the fold test produces inconclusive results is not readily interpretable, but the weight of the evidence suggests that acquisition of the remanence was nearly complete before slumping. Data for the syndepositional slump thus suggest that the characteristic magnetization in nonhematitic Bluewater Creek strata was locked in during or soon after deposition.

Acquisition of isothermal remanent magnetization (IRM) indicates that most samples of the Bluewater Creek Formation contain contributions from low- and high-coercivity phases (Figure 4). AF demagnetization of the IRM suggests that contributions from a high-coercivity phase are more important. The high coercivity but distributed unblocking temperatures of the ChRM of Bluewater Creek strata suggest that fine- to coarse-grained hematite is the main remanence carrier. The high coercivity and low unblocking temperature (100-300°C) of the steep, south directed overprint suggests that this is also carried by fine-grained hematite. Notably, however, many of these rocks lack hematite pigment. Microscopic observations show that small concentrations of hematite as specularite are present in the immediate vicinity of some ferromagnesian and iron oxide detrital grains. The contribution of a low-coercivity phase to the IRM suggests that magnetite is the possible carrier of viscous components of recent origin.

A large percentage (~60-70%) of hematite grains contain ilmenite intergrowths and typically show evidence of transport (e.g., rounding); we infer that these are of detrital origin. These grains are possibly the carriers of the

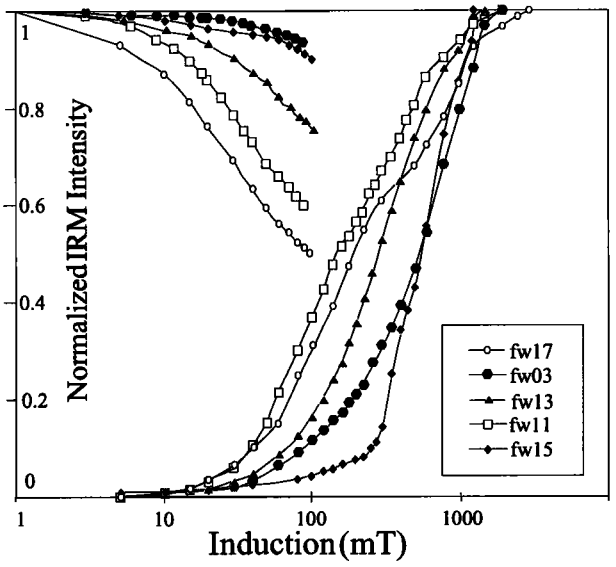


Figure 4. Isothermal remanent magnetization (IRM) acquisition curves and AF demagnetization of the IRM for selected samples of the Late Triassic Bluewater Creek Formation, the Middle Triassic Moenkopi Formation, and the Lower Permian San Andres Limestone.

characteristic magnetization. Some of these grains have been partially corroded and hematite has been remobilized (Figure 5). Hematite is also found at grain boundaries, as alteration rims on ferromagnesian silicate grains, but this occurrence is relatively rare. A common occurrence of hematite (in some specimens making 40 to 50% of all the grains) is as aligned inclusions in detrital mica and chlorite (Figure 5). Some inclusions are lensoidal in shape, others are planar. Inclusions are often truncated at grain boundaries and in some cases they are deformed through compaction. There is almost no evidence of hematite dissolution and no cases of hematite forming around the mica grains except for small stains of hematite pigment in the most oxidized grains. These grains are completely enclosed in calcite cement, and, in some instances, cement crosscuts inclusions. These observations suggest that hematite inclusions formed before deposition of sheet silicate grains; they may thereby contribute to a depositional remanence (DRM). Magnetite is present in other, less altered, ferromagnesian silicate grains. Pigmentary hematite is rare even at the microscopic level and is concentrated in the immediate vicinity of specular hematite grains. A less common occurrence of hematite is as a relatively "late" diagenetic phase in veins that crosscut earlier calcite cement. Late diagenetic fine-grained hematite may carry the steep south directed overprint unblocked between 100° and 375°C.

The mean virtual geomagnetic pole (VGP) derived from sites in hematitic Bluewater Creek strata with high unblocking temperature (>650°C) magnetizations is indistinguishable at a 95% confidence level from that derived from nonhematitic sites; the angular distance between the means is only 3°. The characteristic magnetization of hematitic samples is thus also interpreted as an "early acquired" magnetization.

Some specimens from sites fw9 and fw10 have magnetizations we tentatively interpret to be of normal polarity. The interpretation is tentative because they may also reflect acquisition of a secondary magnetization of uncertain origin. It is unclear whether or not these sites are in continuous stratigraphic sequence, but both are below the McGaffey member. These sites, as well as sites fw8, fw18, and bw3, yield demagnetization results of inferior quality (Table 1; only site fw10 was included in the overall mean). Their polarity is, for the most part, interpreted from endpoint directions. These are poorly resolved north directed and shallow magnetizations (Figure 2g); the magnetic moment remaining after heating to 500°C is near the magnetometer noise level (10^{-9} A m^{-2} kg), but a similar behavior was observed in at least one other specimen from this site. Other samples from this site give east directed magnetizations

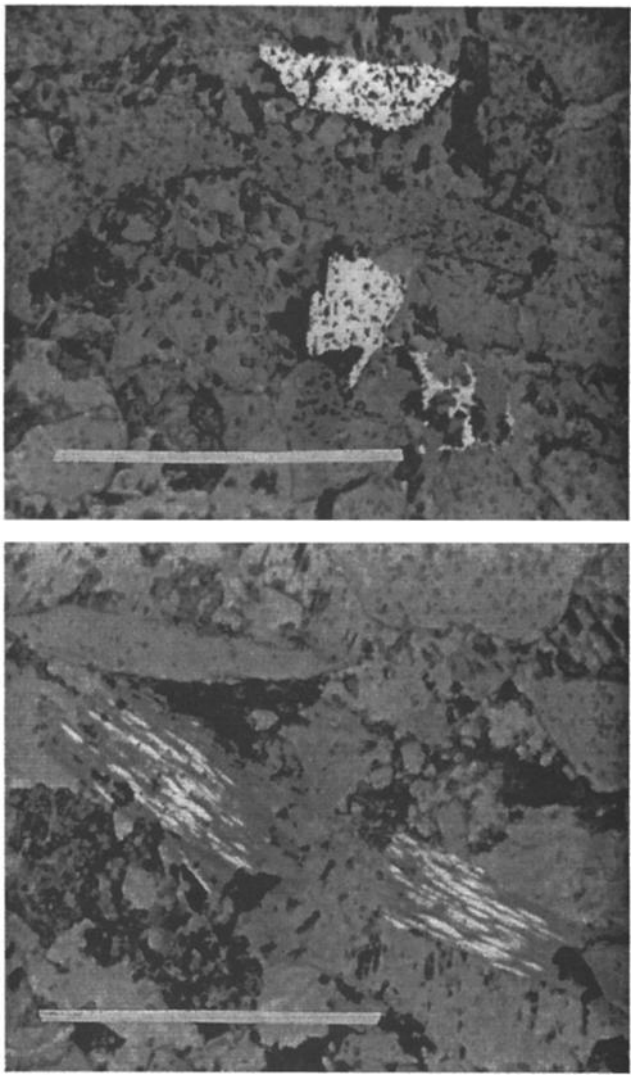


Figure 5. Photomicrographs of common hematite occurrences in sandstone beds of the Bluewater Creek Formation near Fort Wingate. (top) Rounded to subrounded specularite grains showing extensive remobilization and recrystallization of hematite (specularite) at grain margins. Late diagenetic hematite bearing veins are in the lower right corner. (bottom) Inclusions of specular hematite along cleavage plains of chlorite grains. In some of the grains inclusions appear to be isotropic suggesting that some magnetite is also present in similar forms. Scale bars are approximately 40 μm.

Table 1. Paleomagnetic Data and Statistical Parameters

Site	n\nd	In Situ Declination, deg.	In Situ Inclination, deg	k	α95	Tilt Corrected Declination, deg.	Tilt Corrected Inclination, deg	VGP Latitude, deg.	VGP Longitude, deg.
Bluewater Creek Formation, Fort Wingate (35.46°N, 108.54°W)									
fw01†	5\7	139.7	14.9	73.2	9.0	157.3	-3.1	50.1	108.4
fw02†	4\7	149.1	14.9	45.5	13.8	157.6	-12.1	54.2	111.8
fw03†	3\5	144.1	10.7	26.4	24.5	153.2	-7.4	49.7	115.6
fw05†	7\8	151.0	28.0	44.7	9.1	171.1	-12.8	59.9	89.3
fw06	3\6	175.4	3.0	19.3	28.8	175.6	8.6	50.0	78.3
fw07†	3\6	156.7	13.7	45.7	18.4	156.9	-20.6	57.5	117.3
fw08†*	1\6	12.9	6.6			12.7	-2.9		
fw09*	1\5	344.6	-2.9			344.3	-11.6		
fw10	5\8	180.1	5.2	12.4	22.6	179.1	2.1	53.5	73.0
fw11	4\8	188.4	10.4	37.7	15.2	181.3	18.0	43.3	69.6
fw18*	3\7	167.4	18.8	14.0	34.3	168.6	30.5		
Moenkopi Formation, Fort Wingate (35.46°N, 108.54°W)									
fw12	3\5	161.9	-3.8	34.7	21.2	161.9	2.2	49.7	100.2
fw13	2\8	159.2	-20.7			159.1	-15.0	56.2	111.0
fw16	5\7	160.4	3.6	11.2	24	160.5	9.6	45.7	99.9
San Andres Formation, Fort Wingate (35.46°N, 108.54°W)									
fw14	3\4	359.8	54.6	1150	3.6	356.8	49.1		
fw15	3\4	352.8	57.7	1354	3.4	350.4	51.9		
fw17	3\4	156.2	17.8	7.3	50	156.2	23.8		
Bluewater Creek Formation, Bluewater Creek Lake (35.29°N, 108.06°W)									
bw01	6\6	169.0	-11.0	26.4	13.3	169.1	-9.1	57.7	92.6
bw02	7\7	184.0	-0.1	43.5	9.3	184.0	1.9	53.6	65.2
bw03*	2\11	166.8	27.7			166.5	29.6		
bw04	5\8	187.5	0.5	31.3	13.9	187.5	2.5	52.8	59.5
bw05	5\8	181.4	-14.5	18.7	18.2	181.4	-12.5	61.0	69.1
bw07	4\4	168.6	-1.1	39.3	14.8	168.6	0.8	52.8	91.0
Moenkopi Formation, Bluewater Creek Lake (35.29°N, 108.06°W)									
bw8	3\3	176.3	0.4	501.1	5.5	176.3	2.4	53.3	78.1
bw9	5\8	350.8	10.8	11.4	23.6	350.9	8.8	58.0	89.3
bw11	7\8	344.4	10.6	26.5	11.9	344.5	8.7	55.9	100.3
San Andres Formation, Bluewater Creek Lake (35.29°N, 108.06°W)									
bw10	4\9	306.3	15.4	9.8	30.9	308.1	4.4	31.7	139.5
Other Directions									
bw10Hi-T	3\9	337.4	58.0	675.5	4.7	340.0	43.3	70.7	135.7
fw Lo-T	19	163.4	-61.8	32.6	6.0	162.1	-55.9	75.5	170.6
fw17Hi-T	5\8	169.5	-57.4	143.6	6.4	167.5	-51.5	79.1	147.4
Mean Bluewater Creek Formation						k=39.7	A 95=6.7°	55.2	87.5
Bingham statistics				(k 1=-102.7; k 2=-14.3;ov-az=85° ; α13=2.8°; α12=7.6°)					
Mean Moenkopi Formation						k=109.2	A 95=6.4°	53.1	96.3

Here n (nd) is the number of samples used (demagnetized) in mean calculations; VGP is the virtual geomagnetic pole; k and a95 areFisher's precision parameter and radius of the 95% cone of confidence.

† Sites sampled in a syndepositional slump.

* Sites excluded from calculations.

(Figure 2h). It is possible that these are composite magnetizations, reflecting "long-term" acquisition of a chemical remanence [*Larson et al.*, 1982]. If the normal polarity event these rocks record is indeed short, as we have assumed, the east directed magnetizations can be explained as the sum of normal and reverse polarity chemical remanences. Samples from site fw10 are predominantly reverse (Figure 2j), but in two samples a moderate to high coercivity (>30 mT), north directed magnetization (also poorly defined) is removed over the range of temperatures between 350° and 550°C (Figure 2i). The north directed endpoint vectors and the component removed in site fw10 could be interpreted as overprints of uncertain nature. Their magnetizations are too shallow, however, to be a recent overprint. Therefore we tentatively interpret these as normal polarity magnetizations. The late Carnian is characterized by predominantly reverse polarity; normal polarity subchrons are interpreted to be relatively short (e.g., ~ <200 kyr) in duration [*Kent et al.*, 1995]. It is thus possible that sites fw9 and fw10 have somewhat noisy records of one of the inferred short events. They do not provide, however, sufficient information to attempt a reversal test.

The VGPs of 13 of 17 accepted sites in Bluewater Creek strata give a mean pole at 55.2°N, 87.5°W (Table 1). The section of Bluewater Creek Formation studied is about 50 m thick. Even for a relatively fast rate of deposition in a cratonic

setting, deposition of this sequence seems to span a time interval long enough (>500 kyr) to average secular variation. The circular standard deviation of the VGP distribution (12.9°) also suggests a reasonably adequate averaging of secular variation. It appears, moreover, that our collection of data spans at least two reverse chrons and one short normal polarity event. Thereby, despite the relatively small number of accepted sites, the mean represents an accurate estimate of the late Carnian paleopole position.

Because of the sampling scheme and the lack of well-indurated layers, the Bluewater Creek magnetostratigraphy may lack sufficient resolution for rigorous comparison with proposed Triassic polarity sequences. It is permissible that the reverse-short normal-reverse sequence correlates with intervals between zones E8 and E12 of the Newark polarity record [*Kent et al.*, 1995].

A steep negative and south directed magnetization (in situ: D=163.4°, I=-61.8°; tilt corrected: D=162.1°, I=-55.9°) removed at low temperatures during demagnetization is well defined in 19 samples (Figure 3d). Tentatively, we interpret this magnetization as a late Mesozoic to early Cenozoic overprint of diagenetic origin. Similar directions of magnetization have been identified in Permian and Triassic strata in central New Mexico [*Molina Garza et al.*, 1991].

3.2. Middle Triassic Moenkopi Formation and Middle Permian San Andres Limestone.

In specimens of the Moenkopi Formation the ChRM unblocks over a distributed range of temperatures between 300° and 670°C (Figure 6a) or over a narrow range of unblocking temperatures between 625°C and 670°C (Figure 6b). The ChRM is south or north directed and shallow in inclination (Figure 6c). Components of magnetization in the direction of the present dipole field (I=55°) are present in some of the samples. The six Moenkopi sites yield a well-grouped mean pole at 53.1°N, 96.3°E (Table 1). The number of sites involved in the pole calculation is small, but the pole is similar to previously published Middle Triassic data [*Van der Voo*, 1990]. The Moenkopi Formation is one of the best paleomagnetically characterized rock units in North America, and data for the small number of sites reported here are not anomalous. The angle between the mean of normal and reverse directions is 9.2°. The data pass a reversal test with a C classification [*McFadden and McElhinny*, 1990].

Some workers have suggested that remanence in the Moenkopi Formation was acquired soon after or during deposition [*Purucker et al.*, 1980]. Others have shown that remanence carriers are secondary (diagenetic) and that acquisition long postdates deposition [*Larson et al.*, 1982]. Moreover, some stratigraphic intervals of the Moenkopi Formation have been selectively, and pervasively, remagnetized in late Mesozoic time [*Molina Garza et al.*, 1991]. We are not in the position to contribute to this discussion because no field tests are possible for Moenkopi Formation strata in the localities studied here.

San Andres Formation specimens collected near the Permian-Triassic disconformity contain single component, north directed and steep positive magnetizations of high coercivity (>110 mT) and moderately high but distributed unblocking temperature. These samples are highly altered and contain oxidized phases, such as hematite and goethite. Less oxidized samples from site fw17 in the San Andres Formation contain two components (Figures 6d and 6e), a shallow southeast directed magnetization, and a high coercivity and intermediate unblocking temperature magnetization of steep negative inclination and southeast directed declination (Figure 3d). The IRM of these samples is characterized by a large contribution from a cubic phase. For site bw10, also in the San Andres Formation, magnetizations are either of high unblocking temperature (~650°C) and north-northwest directed and steep positive, or of low unblocking temperature (~400°C) and northwest directed and shallow positive. North to northwest directed and steep positive magnetizations in San Andres strata are most likely recent in origin. The steep negative south to southeast directed, high unblocking temperature magnetization of site fw17 is indistinguishable

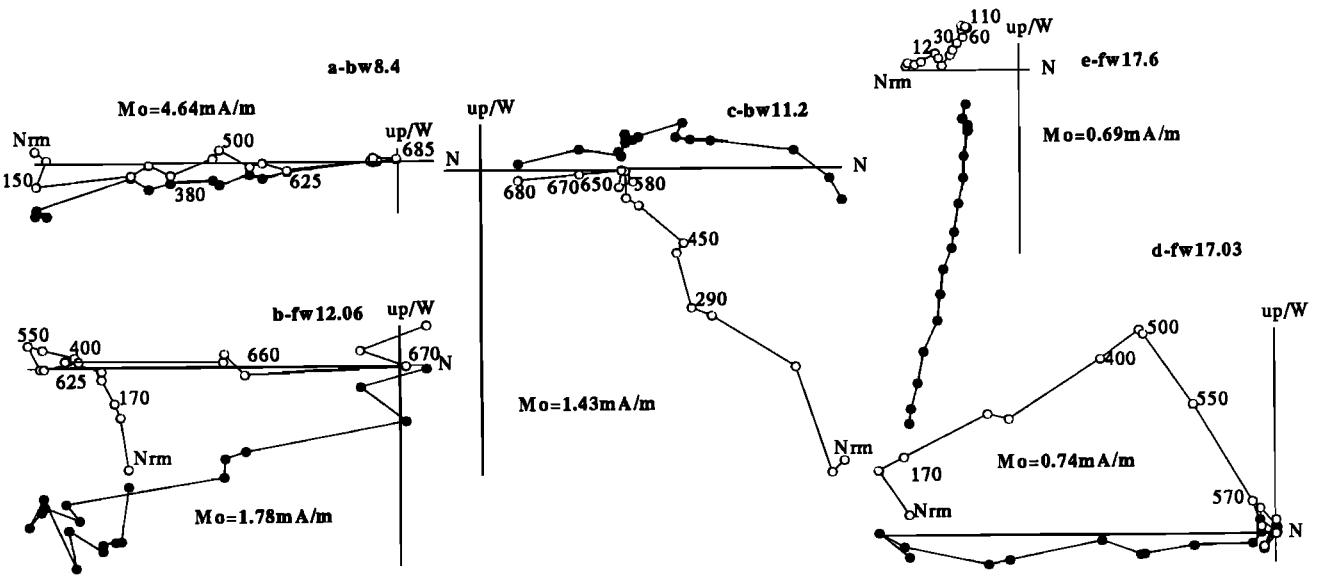


Figure 6. Orthogonal demagnetization diagrams [*Zijderveld*, 1967] of selected specimens of (a-c) Moenkopi Formation and (d, e) San Andres Formation. All diagrams in in situ coordinates. Open (solid) symbols are projections on the vertical (horizontal) plane.

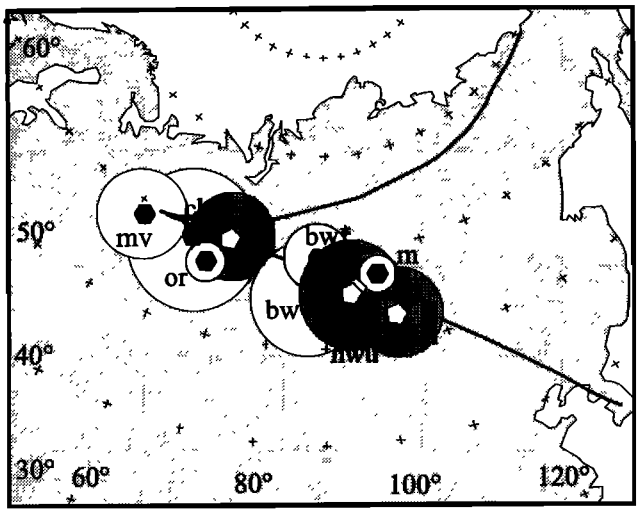


Figure 7. Paleomagnetic poles with 95% confidence regions from Lower Triassic (Smithian) through Lower Jurassic (Sinemurian) strata of the Colorado Plateau (solid symbols) superimposed on the paleomagnetic Euler pole track of *Gordon et al.* [1984]. We include the mean of Moenkopi strata on the CP (calculated using the data of Table 2). CP data are compared with results for the Newark Supergroup (open symbols), interpreted to indicate small to negligible Late Triassic apparent polar wander [*Kent and Witte*, 1993] and a pole interpreted to show evidence of the J-1 cusp in the APWP [*Kodama et al.*, 1994]. The Bluewater Creek pole of this study (bw) is indistinguishable from the mean of equivalent strata in central New Mexico (bw*). The sequence of poles, in ascending stratigraphic order are: m, Moenkopi Formation; bw, Bluewater Creek Formation, western New Mexico (this study); bw*, Bluewater Creek Formation equivalent strata, central Rio Grande rift, New Mexico [*Molina Garza et al.*, 1991]; or, Owl Rock Formation [*Bazard and Butler*, 1991]; ch3, Church Rock Formation [*Kent and Witte*, 1993]; mv, Moenave Formation [*Ekstrand and Butler*, 1989]. The Newark poles are nwl, Newark lower beds [Witte and Kent, 1989]; nwm, middle Newark [*Witte and Kent*, 1990]; nwu, Newark Hettangian zone [*Witte et al.*, 1991]; ncp, Newark Culpeper Basin baked sediments [*Kodama et al.*, 1994].

from the overprint found in Chinle strata and is also interpreted to be of late Mesozoic-early Cenozoic age.

4. Implications for North America APW

The paleomagnetic pole for the Bluewater Creek Formation (55.2°N - 87.5°E) compares favorably (Figure 7) with results for correlative strata in central New Mexico. The lowermost Chinle pole of *Molina Garza et al.* [1991] falls at 60.8°N - 88.9°E. The validity of this pole for interpretations of APW had been questioned because data were collected from localities within or along the margins of the Rio Grande rift, at the eastern edge of the CP. We had previously suggested that this area "was rotating rigidly with the plateau, or it experienced similar sense rotation via an independent deformation mechanism" [*Molina Garza et al.*, 1991, p. 14254] The fact that Early Permian, Middle Triassic, and, now, Late Triassic poles in central New Mexico are indistinguishable from coeval poles for the plateau reinforces that conclusion.

The paleopole for the Bluewater Creek Formation falls to the west of poles for equivalent age strata in the southwestern plains of the craton interior but it is statistically indistinguishable from them (Table 2). Data for the southwestern plains include the late Carnian and early Norian poles for the Santa Rosa, Trujillo, and Garita Creek formations [*Molina Garza et al.*, 1996], and the Bull Canyon Formation [*Bazard and Butler*, 1991], in eastern New Mexico, and the late Carnian Dockum Group pole-A, in west Texas [*Molina Garza et al.*, 1995]. The mean of these poles falls at 55.1°N - 94.0°E, and the angular distance to the Bluewater Creek pole is 3.7°. This result therefore does not support the hypothesis of a "large" rotation of the Colorado plateau.

The angular distance between the new Bluewater Creek pole and the mean of Norian results for the CP is 12.5°, indicating, within statistical uncertainty, that at least that much APW occurred during Late Triassic time. Poles obtained from the stratigraphic succession of Triassic and Lower Jurassic strata on the CP, including the Smithian to Anisian Moenkopi Formation (mean calculated from Table 2), the upper Carnian Bluewater Creek (this study), the mid-Norian Owl Rock [*Bazard and Butler*, 1991], the upper Norian Church Rock [*Kent and Witte*, 1993], and the

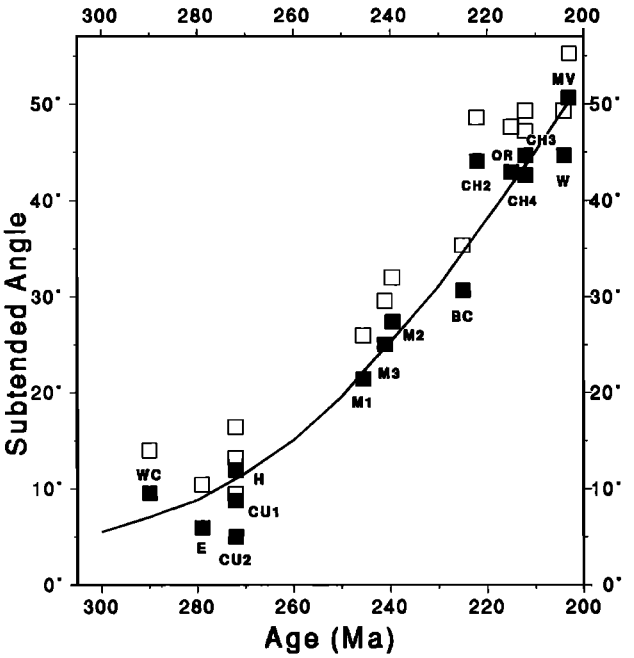


Figure 8. Angular progression of Colorado Plateau paleomagnetic poles along the small circle paleomagnetic Euler pole (PEP) track of *Gordon et al.* [1984] as a function of pole age. We used the PEP in column B of Table 5 in Gordon et al. [1984], which gives a model North American APWP for the Carboniferous though Triassic. The data are shown with (solid squares) and without (hollow squares) a correction for the rotation of the plateau. The correction is made by rotating the plateau poles by 4.6° about an Euler pole at 34°N, 100°W. The solid line is the best fit polynomial through the data from North America craton, which is digitized from Figure 2 of Bryan and Gordon [1990]. Note how well this curve fits the corrected data, including our new Late Triassic data point. Labels are as in Bryan and Gordon [1990] with the addition of our new data, where BW is Bluewater Creek and M3 is our Moenkopi datum. We have also added the Colorado Plateau data given in Table 2 where CH3 and CH4 are the Church Rock data [Kent and Witte, 1993] and OR is the Owl Rock datum [Bazard and Butler, 1991].

Table 2. Summary of Paleomagnetic data for Triassic and Lower Jurassic Nonmarine Strata from West and Southwestern North America

Colorado Plateau	Poles	Craton Platform Interior	Poles
Early Jurassic (205-185 Ma)			
Kayenta Fm.	61.9,74.4	Dockum B overprint (2)	59.7,65.7
Kayenta Fm. (1)	59.0,66.6		
Moenave Fm.	58.2,51.9		
Wingate Fm.	59.0,63.0		
Late Triassic, Norian-Rhaetian (220-205 Ma)			
Chinle, Church Rock Fm. (3)	57.5,63.3	Chinle, Redonda Fm. (4)	(59,77) 58.3,68.9 †
Chinle, Church Rock Fm. (1,10)	(61,64) 59.0,67.0 ‡	Upper Chinle Fm. (5)	57.7,79.1
Chinle, Owl Rock Fm. (1)	56.5,66.4	Chinle, Bull Canyon Fm. (1)	57.4,87.8
Late Triassic, Late Carnian (227-220 Ma)			
Chinle, Bluewater Creek Fm. (6)	55.2,87.5	Popo Agie Fm.	55.5,95.5
Chinle, lower beds (7)	60.5,88.9	Garita Creek Fm. (4)	54.0,86.7
		Santa Rosa-Trujillo Fm. (4)	(53,104) 54.4,97.3 †
		Dockum Group A (2)	56.4,96.3
Middle-Early Triassic (245-235 Ma)			
Upper Maroon Fm	60.0,102.0	Anton Chico Mbr. (4)	48.0,113.5
Upper Moenkopi Fm.	54.8,103.3	Anton Chico Mbr. (8)	43.2,121.4
Lower Moenkopi Fm.	57.2,98.0	Ankareh Fm	50.8,104.6
Moenkopi Fm., AZ	58.0,101.0	Upper Red Peak Mbr.	48.5,104.7
Moenkopi Fm., drillcore	56.0,101.3	Lower Red Peak Mbr.	45.8,121.1
Moenkopi Fm., NM (7)	57.6,100.3	Chugwater Fm.	46.6,113.5
Moenkopi Fm., NM (6)	53.1,96.3	Chugwater Fm.	45.4,115.3
Moenkopi Fm., AZ (9)	58.5,94.8		

References: 1, *Bazard and Butler* [1991]; 2, *Molina Garza et al.*[1995]; 3, *Kent and Witte* [1993]; 4, *Molina Garza et al.* [1996]; 5, *Reeve and Helsley* [1972]; 6, this study; 7, *Molina-Garza et al.* [1991]; 8, *Steiner and Lucas* [1992]; 9, *Steiner et al.*, [1993]; 10, *Reeve* [1975]. Unreferenced poles are from *Van der Voo* [1990]. Poles in parentheses indicate an alternative paleomagnetic result.

† The alternative pole is not corrected by estimated 4° of southward plunge.

‡ Poles recalculated from the original publication (in parentheses).

Hettangian Moenave formations [*Ekstrand and Butler*, 1989] show a smooth progression from east to west (Figure 7) along the paleomagnetic Eule pole (PEP) track of *Gordon et al.* [1984].

For comparison, we consider results for the Newark basin, including the lower Newark [*Witte and Kent*, 1989], middle Newark [*Witte and Kent*, 1990], and upper Newark Supergroup (Hettangian Zone) [*Witte et al.*, 1991], and the Newark-Culpeper baked sediments [*Kodama et al.*, 1994] (Figure 7). The pole progression in the Newark record may be interpreted to indicate abrupt APW in latest Triassic-earliest Jurassic time, but North American data from the high plains of the craton interior suggest a continuous westward progression. Triassic and Early Jurassic poles for southwestern North America (Table 2), ostensibly from on the Colorado plateau, indicate fairly continuous APW as well.

The continuous progression of late Paleozoic to early Mesozoic poles is demonstrated by plotting the angles that poles subtend along the apparent polar wander path (Figure 7), for which we use the small circle APW path proposed by *Gordon et al.* [1984]. If the time interval encompassed by the Carnian and Hettangian (about 225 to 203 Ma in Figure 8) is interpreted to have slow or negligible APW, then extremely rapid APW (>2°/m.y.) would be required for the time periods before and after this interval. Although such an interpretation is permissible, a simpler model with nearly continuous APW at rates of 0.65 to 0.75°/m.y. over the period from the Early Permian to earliest Jurassic, about 290 to 200 Ma, explains the data equally well, without requiring abrupt changes in the motion of North America.

5. Late Triassic data and Colorado Plateau rotation

Kent and Witte [1993] compared the mean of poles for Norian strata of the Chinle Group on the CP with the mean of Norian poles from the Newark Supergroup and the craton interior and concluded that about 13.5° of CP rotation has occurred. If APW was slow during Late Triassic time, as they invoked, then slight differences in the age of the poles would induce negligible bias in their rotation estimate, but that assumption is not supported by data from the CP and the high plains of the Craton interior and should be reexamined. As a first step in such reexamination, we make basic comparisons of mean poles for Norian and Carnian data for the craton with those from the plateau. A far more robust approach is presented by *Molina Garza et al.* [this issue].

We calculated a mean cratonic Norian pole for North America that incorporates the three poles included in the average of *Kent and Witte* [1993] (the Bull Canyon Formation in eastern New Mexico, the Manicouagan impact structure, and the Passaic Formation of the Newark Supergroup) and two additional poles. One of them was omitted by *Kent and Witte* [1993] (the upper Chinle pole of *Reeve and Helsley* [1972]), the other was obtained from the Redonda Formation in the Sangre de Cristo mountains [*Molina Garza et al.*,

1996]. The Norian North America reference pole falls at 58.9°N, 84.3°E (A_{95}=4.3°). For the reference pole, recently published data from the Newark Basin drillcores [*Kent et al.*, 1995] were excluded because their orientation is indirectly determined. The angular distance to the mean of Norian poles for the plateau (Table 2) is 9.9°.

Similarly, we calculated the mean cratonic pole for North America for the Carnian. The pole is located at 52.2°N, 96.7°E (A_{95}=3.2°). In addition to Carnian poles of the Chinle Group off the Colorado plateau (Table 2), the pole calculation includes data for the Dan River Basin [*Kent and Olsen*, 1997], Triassic intrusions in Maine [*Fang and Van der Voo*, 1988], the Fundy Group [*Symons et al.*, 1989], and the lower beds of the Newark Supergroup [*Witte and Kent*, 1989]. The angular distance to the combined Bluewater Creek result, the only Carnian data available for rocks on the plateau, is 6.2°. In total, Carnian and Norian data suggest that rotation of the Colorado plateau is less than 10°. Direct comparison between any two poles from on and off the plateau does not provide, however, the most robust estimate of the amount of CP rotation because each pole has associated uncertainties in its age and position. As other workers have previously concluded, a rotation of 5° about an Euler pole in eastern New Mexico or west Texas brings Triassic poles of the plateau into agreement with the cratonic reference.

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